

High-Level Design (HLD)

RAG-Based Customer Support Assistant

1. System Overview

Problem Statement

Customer support systems are often:

- Slow and manual

- Inconsistent in responses

- Not scalable

Proposed Solution

Design an AI-powered system that:

- Reads PDF knowledge base

- Retrieves relevant information

- Generates accurate responses

- Escalates to human when needed

2. System Architecture

Architecture Diagram (Conceptual)

User (Streamlit UI)

- Query Input

- LangGraph Workflow

- Intent Routing

- RAG Flow OR HITL

- Final Response

3. Components

Document Loader

Loads PDF and extracts text.

Chunking Module

Splits text into smaller chunks (800 tokens with overlap).

Embedding Model

Converts text into vector representations.

Vector Store (ChromaDB)

Stores embeddings for fast retrieval.

Retriever

Fetches relevant chunks using similarity search.

LLM (Groq)

Generates answers based on retrieved context.

Workflow Engine (LangGraph)

Controls execution flow and state transitions.

Routing Layer

Decides whether to answer or escalate.

HITL Module

Handles human intervention when needed.

4. Data Flow

Ingestion Flow

PDF → Load → Chunk → Embed → Store

Query Flow

Query → Retrieve → Generate → Validate → Respond / Escalate

5. Technology Choices

ChromaDB → lightweight vector DB

LangGraph → workflow orchestration

Groq LLM → fast inference

Streamlit → UI

6. Scalability Considerations

Handle large PDFs via chunking

Optimize retrieval using Top-K

Cache repeated queries

Support async processing

Conclusion

The system provides a scalable and intelligent customer support solution using RAG and workflow-based decision-making.